Abdullah U. Mustafa

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Education

(Medical) Physics — Windsor, ON
University of Windsor — Sep. 2023 – Apr. 2027
Honours Bachelor of Science
Minors in Mathematics and in Chemistry

Research Experience

Ultrashort Laser Pulse Measurements — May 2026 – Present
Attosecond Condensed Matter Experiments Lab
Main focus is frequency-resolved optical gating via "Kerr instability amplification" (extreme four-wave mixing). Performed experiments, data acquisition, reconstruction and other analysis, wrote software, and wrote simulations. The long-term goal for this project is to eventually measure damage effects/plasma ionization wrought by KIA in crystals like MgO.

Laser-Induced Breakdown Spectroscopy on Bacterial Cells — Oct. 2023 – Present
Rehse Lab
The main focus is to develop laser-induced breakdown spectroscopy into a quasi-instantaneous diagnostic test for bacterial diseases, such as bacterial meningitis. Led experiments, acquired data, applied chemometric data analysis and machine learning techniques for classification, fixed lasers, and prepared biological samples.

Publications

- *The use of a laser-based diagnostic for the rapid detection and identification of pathogenic bacteria in cerebrospinal fluid*. **Abdullah U. Mustafa**, Lauren Dmytrow, Rachel Chevalier, Simona Brezeanu, Jasmine Fric, Steven J. Rehse. (In preparation.)
- *Kerr instability cross correlation induced nonlinear generation frequency-resolved optical gating*. Nathan G. Drouillard, **Abdullah U. Mustafa**, T. J. Hammond. (In preparation.)

Recent Presentations

- *Establishing the limit of detection of laser-induced breakdown spectroscopy for Escherichia coli in artificial cerebrospinal fluid for the diagnosis of bacterial meningitis*. **Abdullah U. Mustafa**, Lauren Dmytrow, Rachel Chevalier, Jasmine Fric, Simona Brezeanu, Emma Pesce, Katherine Keller, Jack Holland, Will Conlon, Danielle Renard, Faiza Fric, Ayman Younes, Steven Rehse. Canadian Association of Physicists Congress 2026, "Quantitative Biomedical Imaging and Sensing I." June 23, 2026. (Talk delivered by A. Mustafa.) Link to slides.
- *Femtojoule Pulse Reconstruction via Kerr Instability Amplification*. Nathan G. Drouillard, **Abdullah U. Mustafa**, T. J. Hammond. Canadian Association of Physicists Congress 2026, "Ultrafast Phenomena." June 22, 2026. (Talk delivered by T. Hammond.) Link to slides.
- *Differentiation of bacterial species in biomedical specimens using laser-induced breakdown spectroscopy*. Lauren Dmytrow, **Abdullah U. Mustafa**, Rachel Chevalier, Matteo Pontoni, Simona Brezeanu, Jasmine Fric, Katherine Keller, Emma Pesce, Steven Rehse. Canadian Association of Physicists Congress 2026, "Biophysical Approaches to Structure, Function, and Environmental Response." June 22, 2026. (Talk delivered by L. Dmytrow.)

- *LIBS for global health: detecting and diagnosing pathogenic bacteria in human clinical specimens of blood, urine, and cerebrospinal fluid*. Steven J. Rehse, Rachel Chevalier, **Abdullah U. Mustafa**, Lauren Dmytrow, Simona Brezeanu, Jasmine Fric, Emma Pesce. Pacifichem 2025. The International Chemical Congress of Pacific Basin Societies 2025, Honolulu, Hawaii, December 15–20, 2025. (Invited talk delivered by S. Rehse.)

Awards

- NSERC Undergraduate Student Research Award
- Dr. Lucjan Krause Undergraduate Scholarship in Physics
- Outstanding Scholars Award
- President's Entry Scholarship
- Dean's List (3×)
- 1st Place Winner in the Canadian Space Agency Hackathon

Skills

- Optics and Laser Experience
- Python for data acquisition, spectroscopic/chemometric data analysis, machine learning, writing simulations, and writing packages.
- Java, C, and LaTeX (this document).
- Poster and oral talk design and presentation skills.
- 3D modelling and printing with AutoDesk's AutoCAD.

Other Activities

Co-president/Co-founder of Science-Focused Research/Journal Club Jun. 2026 – Present
"Pulse of the Page"
Established and then ran a reading club where participants read and presented science papers of their choosing, in order to practice reading, presenting, and critically analyzing scientific literature.

Teaching Assistant Sep. 2025 – Mar. 2026
Department of Physics
Led (taught, instructed, graded) on pair of introductory physics labs on mechanics and electromagnetism.

Physiotherapy Volunteer Jun. 2025 – Apr. 2026
Hôtel Dieu Grace Healthcare
Volunteered at the physiotherapy floor at a local hospital; assisted patients with their exercises.

Leadership Positions May 2025 – Apr. 2026
"Outstanding Scholars Student Council" and "Physics Club"
Managed outreach and organized events on different student groups.

Mentor Sep. 2024 – Present
"Outstanding Scholars" Mentorship Program
Mentored multiple first-year students through an organized mentorship program.